Lumped-Parameter Modeling and Control for Robotic High-Viscosity Fluid Deposition

William van den Bogert, James Lorenz, Xili Yi, *Graduate Student Member, IEEE*, Albert J. Shih, and Nima Fazeli, *Member, IEEE*

***Abstract*—Robotic high-viscosity fluid deposition plays a pivotal role in various manufacturing applications including adhesive and sealant dispensing, as well as in the additive manufacturing of deformable materials, such as those employed in soft robotics. Uncompensated high-viscosity fluid deposition can lead to poor part quality and defects due to large transient delays and complex fluid dynamics. In this letter, we propose a lumped-parameter flow model and compensation strategies to address significant transient delays and nonlinearity inherent in high-viscosity fluid deposition using a robotic manipulator. Our computationally efficient model is well-suited to real-time control and can be calibrated in minutes. Our compensation strategies leverage an iterative Linear-Quadratic Regulator to compute compensated deposition paths that can be deployed on robotic dispensing systems. These paths can either be deployed offline or corrected live via feedback from our proposed vision-based flow sensor. To validate the effectiveness of our approach, we conducted experiments extruding high-viscosity liquid silicone using a Kuka lbr iiwa robot. Comparative analysis with several baseline protocols demonstrates that our proposed method significantly improves material deposition within desired boundaries.**

***Index Terms*—Additive Manufacturing, Model Learning for Control, Intelligent and Flexible Manufacturing.**

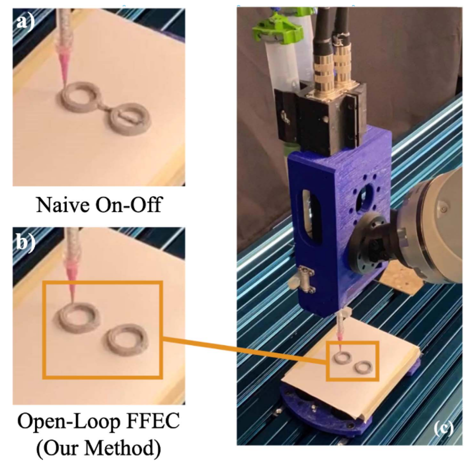

Fig. 1. Robotic 3D printing with model-based flow compensation. A silicone 3D print-head held by a robot arm printing two separated objects. Panel (a) shows the resultant print with naive on-off flow control where we observe significant stringing and part defects due to over-extruded silicone. In comparison, panel (b) shows the result of our open-loop feed forward error compensation (FFEC) where the defect-free objects are separated cleanly. Panel (c) shows the robot 3D printing silicone.

I. Introduction

PRECISE deposition of high-viscosity fluids is a key enabler for a range of applications including sealant deposition and additive manufacturing (AM), specifically the 3D printing of soft robotic components. Such printing of high-viscosity fluids, often curable silicones, is known as direct ink writing (DIW). The fluid's high viscosity aids in maintaining the printed shape, yet without proper compensation, it can lead to printing flaws. Traditional material extrusion (MEX) AM follows a naive control approach, assuming linearity and no time delay between an actuator command and the outlet flow rate. However, when this approach is applied to DIW, the viscosity-induced impedance creates transient effects, resulting in deposition imprecision and issues like the stringing in Fig. 1(a). Common industry protocols include retraction and priming, linear advancing, and coasting, but these are not informed by any model of the deposition process. As a result, significant defects may still occur, as we demonstrate in this letter. Furthermore, the parameters of these solutions are often hand-tuned which can require many iterations.

In this letter, we introduce a framework for modeling and control that enables DIW systems to approach the precision of MEX. Specifically, we contribute:

- a data-driven lumped-parameter model that we hypothesize to be quickly and easily generalizable for most pump-based high-viscosity fluid deposition systems
- a universally implementable model-based open-loop feed-forward error compensation (FFEC) scheme without the need for additional sensors
- a model-based closed-loop control scheme using real-time data from a novel vision-based flow sensor

The target applications of our framework are industries where robotic manipulators are already widely present. Thus, our methods are directly applicable to existing systems in modern manufacturing plants. This practical implementation potential enhances the relevance and scalability of our approach. Using

Manuscript received 25 August 2023; accepted 12 December 2023. Date of publication 4 January 2024; date of current version 18 January 2024. This letter was recommended for publication by Associate Editor Hyun-Jung Kim and Editor Chao-Bo Yan upon evaluation of the reviewers' comments. *(Corresponding author: William van den Bogert.)*

William van den Bogert, James Lorenz, and Albert J. Shih are with the Mechanical Engineering Department, University of Michigan, Ann Arbor, MI 48109 USA (e-mail: willvdb@umich.edu; jplorenz@umich.edu; shiha@umich.edu).

Xili Yi and Nima Fazeli are with Robotics Department, University of Michigan, Ann Arbor, MI 48109 USA (e-mail: yixili@umich.edu; nfz@umich.edu).

Digital Object Identifier 10.1109/LRA.2024.3349931

a Kuka robot, we achieve dynamic extruder movement in 3D space, allowing for dispensing on contoured surfaces, in the same way adhesive is applied at the perimeter of an automotive windshield. This facilitates conformal 3D printing, creating parts with intentional anisotropy or smooth finishes by layering materials on non-planar surfaces. This is crucial for crafting deformable parts to fit complex geometries, such as gaskets or prostheses.

A. Problem Setup and Statement

Our hardware setup can be seen in Fig. 2(a). Two positive displacement pumps (PDPs) force high viscosity uncured silicone through a static mixer and nozzle. Meanwhile, the arm moves this extruder in space to deposit the material in desired locations. Our novel vision-based flow sensor measures the flow rate leaving the nozzle, as feedback for our closed-loop controller. Our PDP system is detailed in Fig. 2(b). We assume the fluids in reservoirs A and B behave identically, and that both motors receive the same input. Here, the control input is the motor speed signal as a function of time $u(t)$, which is proportional to the intended rate of displacement of the progressing cavities in the PDPs. This input produces the nozzle outlet flow rate as a function of time $q(t)$, and both $u(t)$ and $q(t)$ are in units of volumetric flow rate. While naive flow control assumes otherwise, we have observed that $u(t) \neq q(t)\ \forall t$ (see Section IV for examples). For a given time horizon n, our goal is to solve for the trajectory $u(t)$ such that a prescribed $q(t)$ is achieved. We will descretize the time functions as u_k and q_k, defining the trajectories over the horizon n with time steps $k = \{1, \ldots, n\}$. Let $\mathcal{M}$ denote the model mapping the trajectories $\boldsymbol{u}_{n\times 1}$ onto $\boldsymbol{q}_{n\times 1}$:

$$\boldsymbol{q}_{model} = \mathcal{M}(\boldsymbol{u}) \tag{1}$$

$\mathcal{M}$ is an approximation of the true map $\mathcal{R}$ representing the complex nonlinear fluid dynamics. Our first contribution is developing a parametric model $\mathcal{M}^*$ that is i) amenable to real-time control and ii) sufficiently closely approximates $\mathcal{R}$:

$$\mathcal{M}^* = \arg\min_{\mathcal{M}} |\mathcal{R}(\boldsymbol{u}) - \mathcal{M}(\boldsymbol{u})| \tag{2}$$

For open-loop FFEC, we do not have access to $\mathcal{R}$, and instead use the model map $\mathcal{M}$ to find the optimal input trajectory $\boldsymbol{u}^*$ which produces the prescribed output $\boldsymbol{q}_{ref}$:

$$\boldsymbol{u}^* = \arg\min_{\boldsymbol{u}} ||\mathcal{M}(\boldsymbol{u}) - \boldsymbol{q}_{ref}||$$

subject to the constraints of the PDP pump and end-effector motion. Our second contribution is a strategy that leverages our model to compute this optimal open-loop compensation.

For closed-loop control, we have access to a measurement of $\mathcal{R}(\boldsymbol{u})$ after it is subjected to the sensor mapping $\mathcal{S}$:

$$\boldsymbol{q}_{meas} = \mathcal{S}(\mathcal{R}(\boldsymbol{u}))$$

The closed-loop control problem uses the sensor reading to find the optimal input trajectory $\boldsymbol{u}^*$ which produces the prescribed output trajectory $\boldsymbol{q}_{ref}$:

$$\boldsymbol{u}^* = \arg\min_{\boldsymbol{u}} ||\mathcal{S}(\mathcal{R}(\boldsymbol{u})) - \boldsymbol{q}_{ref}||$$

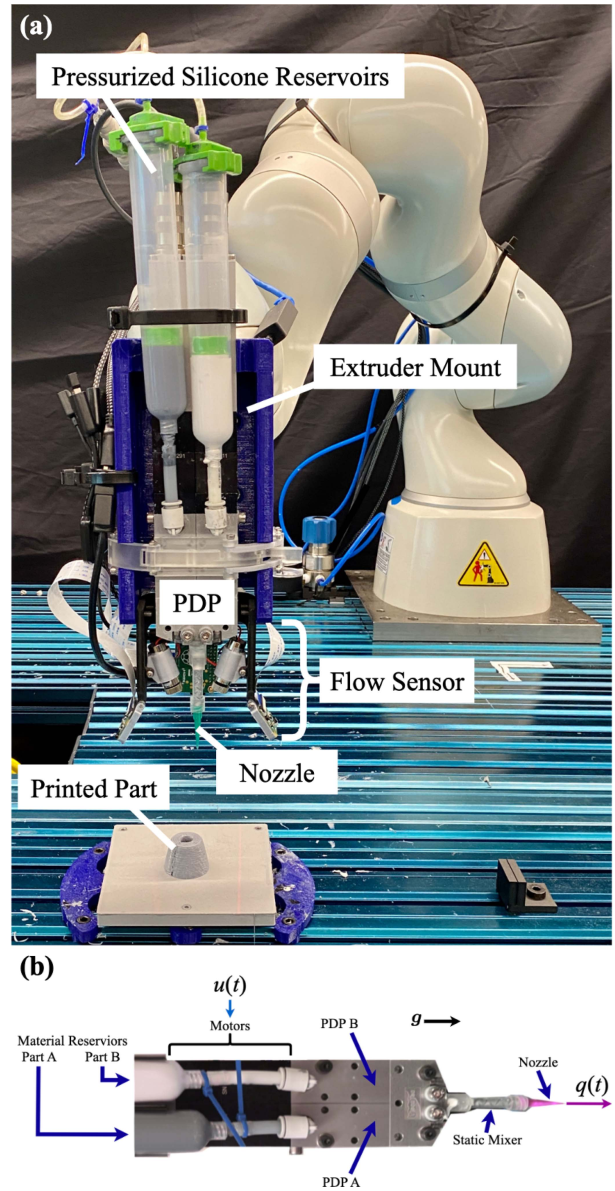

Fig. 2. a) Experiment setup for printing. The whole robotic printing system consists of the robot arm, print-head, and build plate. The print-head includes the extruder, pressurized silicone reservoirs, PDP, flow sensor, and nozzle. The flow sensor is a main contribution of this letter, as it allows use to measure and correct flow rate in real time. b) Positive displacement pump (PDP) system. Two progressive cavity pumps extrude A and B parts of the two-part liquid silicone. A and B parts are mixed via static mixing elements, and mixture is deposited through the nozzle. Here, $\boldsymbol{g}$ represents the direction of gravity.

The third contribution of this work is a strategy that leverages our proposed model alongside real-time measurements to compute this optimal closed-loop control.

II. Related Works

Our work fills a gap in high-viscosity fluid deposition research related to transient flow dynamics in PDP systems for DIW. Existing research on fluid flow behavior in various PDP systems includes control schemes based on analytical solutions to physical laws [1], [2], physics-based simulations [3], [4] [5], and data-driven models [4], [6]. However, these approaches are either too computationally expensive for real-time control or lack consideration of transient effects in PDPs.

Baranovskii and Artemov have proposed generalized solutions for optimal control of nonlinear fluid models based on the Navier-Stokes equations [1], though their analytical solutions demand simplifications specific to their system. This is also the case with the work of Josifovic et al., involving computational fluid dynamics (CFD) to model the step response of a valved pump system [3]. Hapanowicz's flow-resistance model [2] is only applicable to quasi-steady cylindrical flows and does not account for transient pump dynamics.

To control the fluid flow of PDP systems, researchers paired simplified models with closed-loop controllers and numerical solving methods in place of analytical physics-based solutions. Froehlich and Kemmetmüller propose a computationally efficient control [7], but it is designed only for injection molding systems. Wang et al. combine pressure sensors and PID control to correct flow behavior [8]. Numerical solution methods have also been proposed for flow control of screw [9], helical gear [10], and peristaltic pump systems [11]. Lumped-parameter models have been used to compensate pulsing fluid dynamics generated by variable displacement pumps [4] and vane pumps [6], but these models are not generalizable to all PDP types.

Models for progressive cavity pumps are relatively unexplored in academic research. Fisch et al. [12] and Franchin et al. [13] note a strong improvement in precision when a progressive cavity pump is used instead of other PDPs. As a result, research on fluid interaction with progressive cavity PDPs has been limited to basic feedback control [14] and CFD methods to model specific behaviors of the progressive cavity pump system, including wear over time [5], backflow [15], as well as inertial effects and viscous losses [16]. While these models predict internal unsteady flow in progressive cavity systems, they lack predictive capability of the entire dispensing system, which is required for our robotic dispenser.

To bridge these research gaps, we propose a model of the fluid system dynamics that can predict transient effects of high-viscosity fluid flows in PDP systems, which we then pair with both open-loop and closed-loop control paradigms to compensate for deposition errors during deposition.

III. Methods

Our approach is composed of both modeling and control. We will first introduce our modeling framework and then both our open-loop FFEC and closed-loop control strategies.

A. Modeling of High-Viscosity Fluid Deposition

Our goal is to design $\mathcal{M}$ such that it i) is amenable to real-time control and ii) sufficiently closely approximates $\mathcal{R}$ in (2), which represents the complex nonlinear fluid dynamics and interaction with both pump and mixer. We model the entire system as a 3 DOF coupled linear dynamic system, presented as the mass-spring-damper network in Fig. 3. Our model is a representation of fluid flow impedance in the form of elasticity k_j, damping c_j,

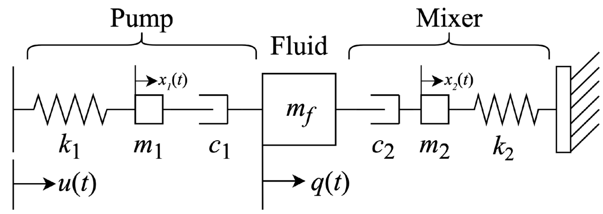

Fig. 3. High-viscosity fluid, pump and mixer are simplified to a 3 ° of freedom coupled linear dynamical system. Our model is a representation of fluid flow impedance in the pump ($j = 1$) and mixer ($j = 2$) in the form of elasticity k_j, damping c_j, and inertia m_j, as well as inertia m_f of the fluid. The "positions" x_1, x_2, q of the "masses" m_1, m_2, m_f are the degrees of freedom that form the state vector $\boldsymbol{x}$ alongside their time derivatives.

and inertia m_j within both the pump ($j = 1$) and mixer ($j = 2$), as well as inertia m_f of the fluid.

With the initial state of the system, our map $\mathcal{M}$ introduced in (1) takes the form of a dynamics function f that predicts the next state $\boldsymbol{x}_{k+1}$ using the current state $\boldsymbol{x}_k$ and input u:

$$\boldsymbol{x}_{k+1} = f(\boldsymbol{x}_k, u_k)$$

The state vector $\boldsymbol{x}$ for the dynamics is defined using the 2 internal degrees of freedom x_1 and x_2 and the output q:

$$\boldsymbol{x} = \begin{bmatrix} x_1 & x_2 & q & \dot{x_1} & \dot{x_2} & \dot{q} \end{bmatrix}^\top$$

We can represent this system in discrete state-space form:

$$\boldsymbol{x}_{k+1} = \mathbf{A}\boldsymbol{x}_k + \mathbf{B}u_k$$
$$q_k = \boldsymbol{y}_k = \mathbf{C}\boldsymbol{x}_k$$

where the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are as follows:

$$\mathbf{A}_s = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -\frac{k_1}{m_1} & 0 & 0 & -\frac{c_1}{m_1} & 0 & \frac{c_1}{m_1} \\ 0 & -\frac{k_2}{m_2} & 0 & 0 & -\frac{c_2}{m_2} & \frac{c_2}{m_2} \\ 0 & 0 & 0 & \frac{c_1}{m_f} & \frac{c_2}{m_f} & -\frac{c_1+c_2}{m_f} \end{bmatrix}$$

$$\mathbf{A} = \mathbf{I}_{6\times 6} + \mathbf{A}_s \Delta t$$

$$\mathbf{B} = \begin{bmatrix} 0 & 0 & 0 & \frac{k_1}{m_1} & 0 & 0 \end{bmatrix}^\top \Delta t$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

The controllability matrix of this system is not full rank, meaning all possible states $\boldsymbol{x} \in \mathbb{R}^6$ in cannot be reached. Since we are only trying to control the output q, we need only evaluate the output controllability matrix $\mathcal{C}_q$:

$$\mathcal{C}_q = \begin{bmatrix} \mathbf{CB} & \mathbf{CAB} & \mathbf{CA}^2\mathbf{B} & \mathbf{CA}^3\mathbf{B} & \mathbf{CA}^4\mathbf{B} & \mathbf{CA}^5\mathbf{B} \end{bmatrix}$$

For our system, this matrix is a single nonzero row, and thus has full row rank, indicating output controllability [17]. Output controllability is maintained if $\mathbf{C}$ is amended with an additional 1 in the sixth column, meaning the flow rate q and $\dot{q}$ are simultaneously controllable to any $\boldsymbol{y} \in \mathbb{R}^2$.

Given measured data from the real system, the parameters are found as the solution to a least-squares optimization. The cost function c is a 2-norm loss for a given time horizon n as a function of the parameters $\boldsymbol{\phi}$, where the data is sampled with the same period as the simulation time-step Δt:

$$c(\boldsymbol{\phi}) = \sum_{k=1}^{n} (q_{k,model}(\boldsymbol{\phi}) - q_{k,meas})^2$$
$$\boldsymbol{\phi} = \left[\begin{array}{ccccccc} k_1 & c_1 & m_1 & m_f & k_2 & c_2 & m_2 \end{array}\right]^\top \quad (3)$$

Where $q_{k,meas}$ is the measured data for the time with index k and $q_{k,model}$ is the modeled flow rate for the time with index k. For each iteration of optimization i the model is solved and the parameters are updated according to the gradient of $c(\boldsymbol{\phi})$:

$$\boldsymbol{\phi}_{i+1} = \boldsymbol{\phi}_i - h\frac{\partial c(\boldsymbol{\phi_i})}{\partial \boldsymbol{\phi}_i}$$

where h is an adjustable increment. We stop this optimization once the cost changes by less than 0.1% between iterations.

B. Open-Loop Control of High-Viscosity Fluid Deposition

With this model in hand, we desire the input profile u for a given time horizon such that the nozzle flow rate tracks the desired q. We construct a quadratic cost function b:

$$b_k = (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref})^T \mathbf{Q} (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref}) + \boldsymbol{u}_k^T \mathbf{R}_k \boldsymbol{u}_k \quad (4)$$

where $\boldsymbol{x}_k$ is the current state, u_k is the current input, $\boldsymbol{x}_k^{ref}$ is the desired state at the time with index k. $\boldsymbol{x}_k^{ref}$ contains the set-point flow rate trajectory q_k, and identical trajectories for the internal states $x_{1,k}$ and $x_{2,k}$, though we will weight the internal state tracking far less than the flow rate tracking.

In our specific case, where our input only has a single degree of freedom, $\mathbf{R}_k$ and $\boldsymbol{u}_k$ reduce to scalars, (4) resolves to:

$$b_k = (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref})^T \mathbf{Q} (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref}) + R_k u_k^2$$

where $\mathbf{Q}$ is a diagonal matrix containing the weightings for tracking error of the 6 states. Since we only are trying to track q, we provide the following diagonal where $\xi \gg \delta$:

$$\mathbf{Q}_{6\times 6} = diag(\delta, \delta, \xi, \delta, \delta, \delta)$$

With the linear dynamics of our model, the cost function defined in (4) could ordinarily be minimized analytically within the bounds of linear quadratic regulator (LQR) theory. However, we have observed that the reference trajectory is tracked more effectively in reality if the weighting on actuation effort R_k is significantly reduced ($r_1 < r_2$) once the input u_k falls below a certain negative value u_{th}, when the pumps are run backwards in an effort to rapidly shutoff the output flow:

$$R_k = \begin{cases} r_1 & u_k < u_{th} \\ r_2 & u_k \geq u_{th} \end{cases}$$

Since the coefficient of actuation effort within the quadratic cost changes depending on the magnitude of the input, we use an iterative LQR (iLQR) solver [18] for this optimization problem. Each iteration i occurs in two parts. The first is a backward pass in which the controller gains $\mathbf{K}_k$ are computed using the value function $\mathbf{V}_k$ at the next time step, which is then updated for the current time step:

$$\mathbf{K}_k = (R_k + \mathbf{B}^{*T}\mathbf{V}_{k+1}\mathbf{B}^*)^{-1}\mathbf{B}^{*T}\mathbf{V}_{k+1}\mathbf{A}^*$$
$$\mathbf{P}_k = \mathbf{A}^* + \mathbf{B}^*\mathbf{K}_k$$
$$\mathbf{V}_k = \mathbf{Q}_k^* + \mathbf{K}_k^T R_k \mathbf{K}_k + \mathbf{P}_k^T \mathbf{V}_{k+1}\mathbf{P}_k$$

where $\mathbf{A}^*$, $\mathbf{B}^*$, and $\mathbf{Q}_k^*$ are the homogenized versions of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{Q}$ respectively:

$$\mathbf{A}^* = \begin{bmatrix} \mathbf{A} & \mathbf{0}_{6\times 1} \\ \mathbf{0}_{1\times 6} & 1 \end{bmatrix}, \quad \mathbf{B}^* = \begin{bmatrix} \mathbf{B} \\ 0 \end{bmatrix}$$
$$\mathbf{Q}_k^* = \begin{bmatrix} \mathbf{Q} & \mathbf{Q}(\boldsymbol{x}_k - \boldsymbol{x}_k^{ref}) \\ (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref})^T\mathbf{Q} & (\boldsymbol{x}_k - \boldsymbol{x}_k^{ref})^T(\boldsymbol{x}_k - \boldsymbol{x}_k^{ref}) \end{bmatrix}$$

The second part is a forward pass in which the input u_k is updated according to the controller gains, and the state $\boldsymbol{x}_k$ is updated according to the dynamics with the input:

$$u_k^{i+1} = u_k^i + \mathbf{K}_{k-1}(\boldsymbol{x}_{k-1}^{i+1} - \boldsymbol{x}_{k-1}^i)$$
$$\boldsymbol{x}_{k+1}^{i+1} = \mathbf{A}\boldsymbol{x}_k^{i+1} + \mathbf{B}u_k^{i+1}$$

We stop the controller optimization if the cost changes by less than $\epsilon = 0.1\%$ between iterations.

C. Closed-Loop Control of High-Viscosity Fluid Deposition

To close the loop on flow rate control, we develop a novel vision-based sensor to measure the just-printed bead. Our sensor projects a laser line from a diode onto the printed bead behind the forward-traveling nozzle, and observing the distortion of the line from a Raspberry PI (RasPI) camera.

At time index k, the sensor directly measures the cross sectional area in pixels of the bead in the camera frame $A_{p,k}$. We assume $A_{p,k}$ scales linearly to the true cross sectional area in physical units $A_{b,k}$ by proportionality constant $K = A_{p,k}/A_{b,k}$. Through a control volume analysis of the recently printed bead, we represent flow rate q_k as a function of both area $A_{p,k}$ and nozzle velocity v_k:

$$q_k = \frac{1}{K}A_{p,k}v_k \quad (5)$$

With two independent variables, this function can be represented as a surface. We identified K as the solution to another least-squares optimization, where the surface defined by (5) is fit to real data. We collected data at various steady state flow rates (>10 s after step-up command) and velocities, and measured $A_{p,k}$ with the sensor. This measurement involves two trained segmentation models for the nozzle and laser line to be masked from the RasPI camera feed respectively. The sensing procedure for finding $A_{p,k}$ is visualized in Fig. 4.

The flow sensor we introduce in this letter provides the unique ability to perform closed-loop control of fluid deposition. The controller from Section III-B is used, but it is continually solved over a horizon T_{CL} that follows the current time, initialized with an estimation for the current flow rate.

After acquiring the sensor reading, we apply a Kalman filter to generate the flow rate estimations q_{init} and $\dot{q}_{init}$. The controller from Section III-B is given an initial state-vector and a set-point

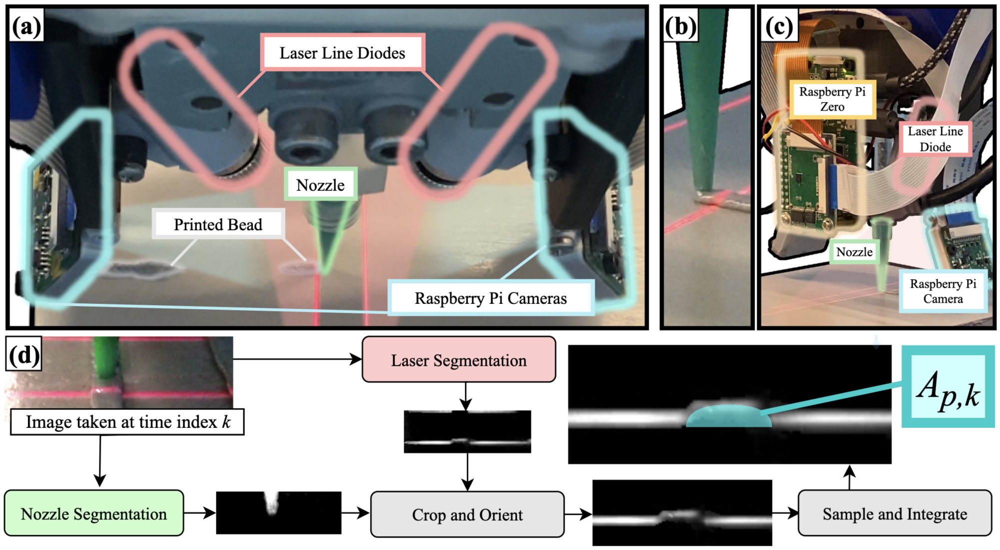

Fig. 4. Novel flow sensor detail and bead area measurement visualization. a) Top view of sensor, showing laser line diodes and cameras underneath the PDPs. b) Close-up of laser line distortion caused by the printed bead. c) Rear view of sensor, showing the Raspberry Pi Zero that collects and sends image data over the ROS network. d) Visualization of bead area measurement. Segmentation of nozzle is used to locate the center of the image (where the printed bead should be) and the laser segmentation is cropped accordingly. The mean location of brightness of each column in the laser segmentation is used to sample the curve of the laser distortion, which is discretely integrated along the horizontal axis to return the bead area in pixels A_{px}.

trajectory q_{ref} for the horizon T_{CL}:

$$\boldsymbol{x}(0) = \begin{bmatrix} q_{init} & 0 & q_{init} & 0 & 0 & \dot{q}_{init} \end{bmatrix}^T$$
$$\boldsymbol{x}_{ref} = \begin{bmatrix} q_{ref} & q_{ref} & q_{ref} & 0 & 0 & 0 \end{bmatrix}^T$$

Once the optimal input u_k is found, its initial value u_0 is sent to the pumps. To mitigate extreme discontinuities this input to the pumps is real-time filtered using a simple moving average of the previous z samples (SMA_z). We wrote the program such that the controller solves every time the sensor has a new measurement, which occurs after each period equal to Δt_{sens}. The assumption for this procedure is that the duration required for solving the controller is less than Δt_{sens}.

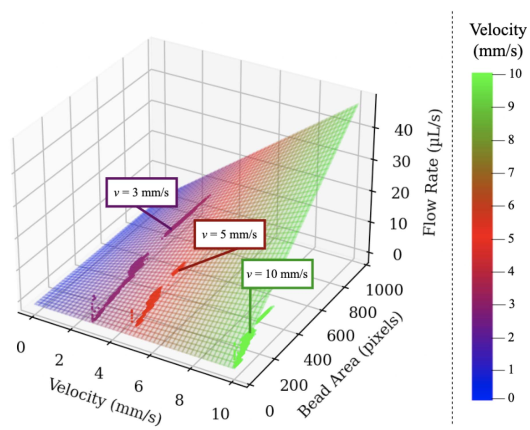

Fig. 5. Sensor calibration surface $q(A_p, v)$ described in (5), with $K = 213.7 px/mm^2$, alongside data collected at nozzle speeds 3, 5, and 10 mm/s.

IV. Results

A. Experimental Setup and Data Collection

Our experimental setup can be seen in Fig. 2 and detail of the flow sensor can be seen in Fig. 4. We used a 7DOF Kuka LBR 14 R820 robot arm equipped with a customized print-head alongside an adjustable build plate fixed to the base of the world frame. The customized print-head consists of the extruder mount, pressurized reservoirs, PDPs, mixer, nozzle, and flow sensor. The silicone material used for deposition is DOWSIL$^{\text{TM}}$ 121. The stepper motors in the PDPs are controlled by a Raspberry Pi 4 Model B using integration with Python and ROS, while the sensor operates through a Respberry Pi Zero that communicates with the same ROS network. The silicone reservoirs are connected to a high-pressure air source limited at 100 psi by a pressure regulator.

We collected all flow rate data presented in this section with our vision-based flow sensor, described in Section III-C. Fig. 5 shows the data for sensor calibration and the surface defined by (5) with the calibrated parameter $K = 213.7$ px/mm^2.

B. Model Training

To implement the model training algorithm described in Section III-A, we used values of $h = 0.1$ for the step increment, and $\Delta t = 0.01$ s for the simulation time step. The data seen in Fig. 6 was collected by the sensor over 4 trials.

We trained the model on an input command trajectory consisting of 10 square pulses of different magnitudes, both negative and positive, as seen in Fig. 6. This was to anticipate the necessary distribution for compensating step-up and step-down flow rate trajectories at the nozzle. The optimization returned the parameters shown in Table I.

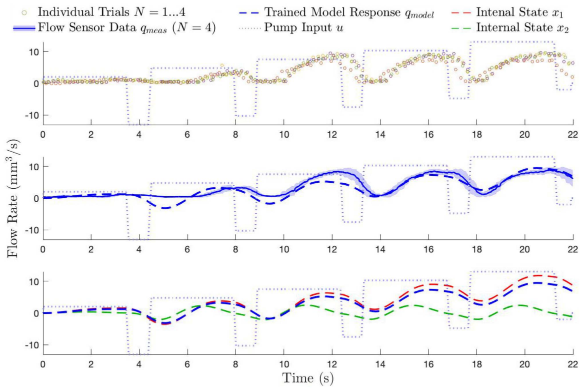

Fig. 6. Lumped-parameter fluid deposition model was trained on 4 sets of data collected live from the flow sensor while the pumps were given the input u over a horizon of 22 seconds. Due to the real non-linear and specifically non-Newtonian nature of the silicone, a linear model cannot accurately predict across the entirety of a wide range of flow rate magnitudes. Additionally, while negative flow is predicted by the model, the sensor has no means to measure it, though it may truly occur. The internal state response is also shown.

TABLE I
PARAMETERS FOUND AS A RESULT OF MODEL TRAINING

k_1	c_1	m_1	m_f	k_2	c_2	m_2
1.77	18.81	1.02	1.78	9.87	5.33	1.11

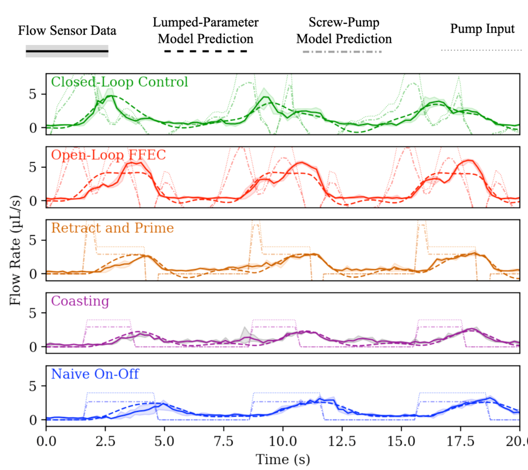

Fig. 7. Prediction of unseen data by the lumped-parameter model. With the parameters from trained model, the model is able to predict unseen data, consisting of the flow rate input to the pumps and output measured by the flow sensor from the pulse validation experiment. Our open-loop FFEC provided $u \in [-16.0, 12.7]$ and our closed-loop controller provided $u \in [-6.0, 11.8]$. Our prediction is compared with the screw pump model from Liu et al. [19].

With these parameters defined, the model can predict the data used in training as shown in Fig. 6. Fig. 7 shows the ability of the trained model to predict unseen data, compared with an existing screw-pump model [19].

C. Control and Compensation

We tested both our open-loop FFEC and our closed-loop controller against 2 industry protocols, for tracking both "pulse" and "dip" set-point flow rate trajectories. The pulse task demonstrates the ability of the flow rate to turn on and off, especially repeatedly, as it may while printing small areas during layered 3D printing. The dip task demonstrates the ability of the flow rate to drop briefly before returning to a nominal value, as it should to avoid bulging during deceleration of the nozzle while dispensing in a tool path that contains sharp turns. All data was collected via the sensor described in Section III-C. For all tests, the desired flow trajectory, pump input trajectory, and cartesian trajectory of the robot were sampled at a period of $\Delta t_{sens} = 0.2$ s for commanding the arm and PDPs. We chose parameters for industry protocols based on typical values for MEX.

To implement the open-loop FFEC solver described in Section III-B, we used values of $\xi = 100$ and $\delta = 0.01$ for the tracking weight, and $r_1 = 20$, $r_2 = 200$, $u_{th} = -2$ mm $^3/$ s for the actuation effort weighting. The desired flow trajectory, generated with a time step of $\Delta t_{sens} = 0.2$ s, was linearly interpolated to $\Delta t = 0.001$ s for solving.

Our closed-loop controller described in Section III-C was implemented with values of $\xi = 100$ and $\delta = 0.01$ for the tracking weight and $r_1 = r_2 = 80$ for the actuation effort weighting, $T_{CL} = 5.0$ s for the solving horizon, $\Delta t = 0.06$ s for the simulation time step, $z = 2$ for the SMA on u, and the stop condition for the optimization was changed to $\epsilon = 2\%$. Table III compares this controller on the pulse task to others with varied select parameters.

For the pulse task, we commanded a series of 3 square pulses with a magnitude of 4 μL/s. The industry protocol applied to this task were retract/prime and coasting. For retract/prime, at each rising edge 10 μL/s is commanded for 0.4 s, and at each falling edge –10 μL/s is commanded for 0.4 s. For coasting, the falling edge is commanded to the pumps 1 s earlier than the desired output. The pulse task was tested 3 times, for a total of 9 pulses sent to each control method. The mean of all 9 output pulses is shown in Fig. 8, for each of the 4 control methods alongside the naive control, in which the set-point is used as $u(t)$. The mean-absolute-error (MAE) between the trajectories produced by each method and the set-point can be seen in Table II.

For the dip task, we commanded the printer to extrude at 4 μL/s for a period of time before briefly shutting off and returning to 4 μL/s. The industry protocols applied to this task were linear

TABLE II
MAE RESULTS FOR PULSE TASK

	MAE	% Improvement
Closed Loop Control	0.92 ± 0.28	46
Open Loop FFEC	0.81 ± 0.20	52
Retract and Prime	1.26 ± 0.16	25
Coasting	1.52 ± 0.17	10
Naive On-Off	1.69 ± 0.21	0

TABLE III
VARIED CLOSED-LOOP CONTROLLERS (CLC) FOR PULSE TASK

CLC	$r_1 = r_2$	T_{CL}	ϵ	u Filter	MAE
1	80	5.0 s	0.02	SMA_2	0.92 ± 0.28
2	70	4.0 s	0.03	SMA_3	0.98 ± 0.39
3	170	1.4 s	0.03	SMA_4	0.99 ± 0.31

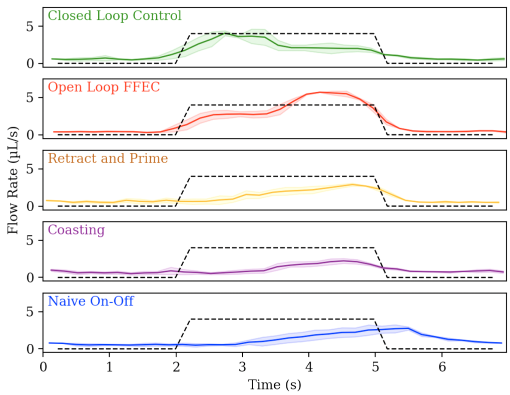

Fig. 8. Results of pulse test. The dotted line represents the set-point.

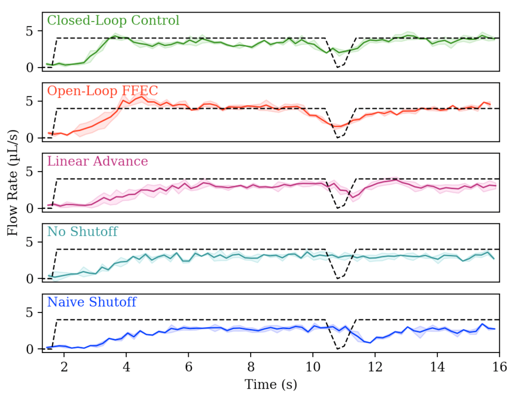

Fig. 9. Results of dip test. The dotted line represents the set-point.

TABLE IV
MAE Results for Dip Task

	MAE	% Improvement
Closed Loop Control	0.88 ± 0.30	49
Open Loop FFEC	0.82 ± 0.34	53
Linear Advance	1.40 ± 0.41	20
No Shutoff	1.25 ± 0.32	28
Naive Shutoff	1.74 ± 0.22	0

advance and no shutoff. We consider no shutoff an industry protocol because often no flow compensation is applied for sharp corners in the tool path. For linear advance, the pump motors decelerate to 0 μL/s early before accelerating to 7 μL/s briefly, then returning to 4 μL/s. No shutoff involves a constant 4 μL/s commanded to the pumps. We tested the dip task 3 times, and the mean of the 3 responses can be seen in Fig. 9, for each of the 4 control methods alongside the naive shutoff control, in which the set-point is used as $u(t)$. The MAE between the trajectories produced by each method and the set-point can be seen in Table IV.

We also evaluated the pulse task using observations of material placement using a top view of the print bed after the deposition test. The 3 trials were done again and the results were photographed, automatically segmented to a binary mask using MATLAB R2020b, and compared with a grayscale mask of the desired material placement using binary-cross entropy (BCE)

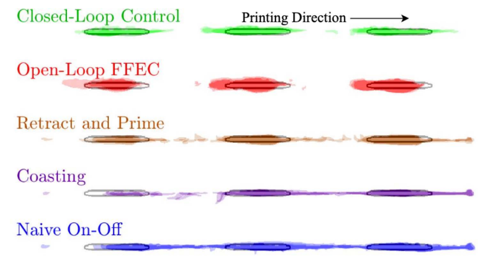

Fig. 10. Visualization of material placement masking that was used for the binary cross entropy analysis. The black outlines indicate the boundary of desired material placement, while the colored areas represent resulting material placement. Higher opacity indicates placement during multiple trials.

TABLE V
BCE Results for Pulse Task

	BCE/10^4	% Improvement
Closed Loop Control	2.5 ± 0.3	49
Open Loop FFEC	2.8 ± 0.5	43
Retract and Prime	3.2 ± 0.7	35
Coasting	4.8 ± 0.2	2
Naive On-Off	4.9 ± 0.3	0

loss as an evaluation method. This grayscale mask has a small dilation blur to less harshly penalize material slightly outside of the intended boundary. The masking comparisons and BCE results can be seen in Fig. 10 and Table V.

D. Application to 3D Printing

We tested our method and the industry protocols while 3D printing two hollow truncated cones, simultaneously. Our method shows improvement in the seam that runs along the side of the cone, which might be critical for preventing leaks in pneumatic soft robots, or providing a smooth surface for gripper fingers. The qualitative results can be seen in Fig. 11.

V. Discussion and Limitations

In this letter, we show that the identification of 7 parameters that define our model allows us to apply it to both open-loop or closed-loop approaches of correcting flow errors. We validated our approaches with deposition tests using our robotic 3D printer. Fig. 11 shows that the naive on-off pump input leads to unintended deposition between two 3D printed cones, which are separated cleanly when open-loop FFEC is applied. For closed-loop flow control during AM, our flow sensor must be modified to align the laser/camera pairs with the direction of nozzle travel. Beyond AM, Fig. 10 shows both our methods enhance deposition precision along linear trajectories, which is applicable for robotic deposition of sealant and adhesive. All evaluations we conducted on the pulse and dip tasks show an improvement of over 40% using our methods.

The versatility of our linear model hypothetically extends to any material or PDP, provided it is retrained on the system that will be used during the final application. Though our results obtained from a single pump/material combination are not a test of this hypothesis, we hold this hypothesis as our mass-spring-damper network contains salient features of a

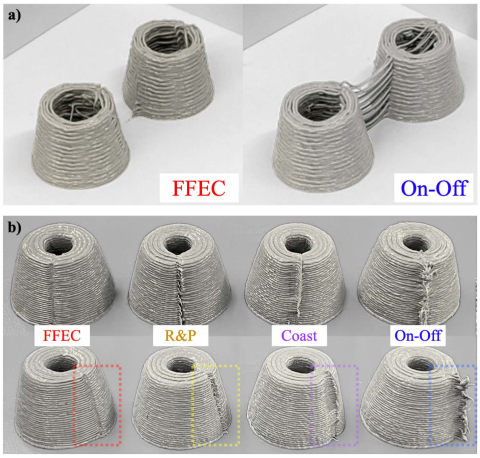

Fig. 11. 3D printing results, where a) and b) represent separate experiments printing differently sized cones. a) Our method (FFEC) compared with the naive on-off protocol, showing the positioning of the two cones being printed simultaneously. b) An isolated single cone from each two-cone experiment conducted for the methods discussed in this letter. Note the roughness present in the seam resulting from the naive on-off protocol, and the gaps present in the seam resulting from the industry protocols. When using FFEC, the seam is blended smoothly. .

pump/material/mixer system. DOWSIL^{TM} 121 is similar to 3D printing silicones that exhibit viscosities between 100 and 1000 Pa-s, which is common for our target applications [20]. While material properties can change with ambient conditions such as temperature, this work assumes training and testing occur in the same environment; this could be considered a limitation. A physics-based model would be beneficial in adapting the controller to changing ambience. While our model's predictions are satisfactory for this letter's intent, Figs. 6 and 7 show predictions that are missing some measured features, likely due to sensor and/or model inaccuracy. Our linear model lacks wholly predictive capabilities for non-Newtonian silicone behavior. A more expressive model of the fluid dynamics could mitigate eccentricities of predictions seen in Fig. 7, especially since iLQR is compatible with nonlinear systems [18].

One motivation to keep our model computationally inexpensive is to allow further improvements of closed-loop control, regardless of the type of sensor or control scheme used. The flow sensor and closed-loop control scheme used in this letter both require a significant amount of processing time, which has limited us to a loop time of 0.2 s. This is perhaps why closed-loop control manages to consistently perform just under open-loop FFEC according to Tables II and IV. While the closed-loop control must be conducted in real-time, the FFEC can optimize the input trajectory indefinitely before it is deployed. However, according to Table V and Fig. 10, closed-loop control outperforms FFEC. This discrepancy is likely due to differences in how the data are registered to the desired result. While this may suggest that it makes little difference whether our closed- or open-loop solutions are used, closed-loop control has unique advantages that stem from its reliance on real-time observations, potentially outperforming FFEC on unseen systems. We can expect the FFEC to underperform on such systems, where the mixer, material, or pump may have changed from training. A higher frequency closed-loop control scheme coupled with a more accurate sensor could better eliminate errors in real-time. Additionally, the nozzle trajectory can be optimized alongside the flow rate so that the transient delay still present in the corrected output (see Figs. 8 and 9) will not be a permanent barrier to precise material placement. Through this we will achieve even further improvements than those demonstrated in this letter.

References

[1] E. S. Baranovskii and M. A. Artemov, "Existence of optimal control for a nonlinear-viscous fluid model," *Int. J. Differ. Equ.*, vol. 2016, Jun. 2016, Art. no. 9428128.

[2] J. Hapanowicz, "Two-phase liquid–liquid flow in the aspect of reduction of pumping power of hydrophobic substances with high viscosity," *Energies*, vol. 14, no. 9, 2021, Art. no. 2432.

[3] A. Josifovic, J. Corney, and B. Davies, "Valve dynamics in multi-cylinder positive displacement pump model," in *Proc. IEEE Int. Conf. Adv. Intell. Mechatron.*, 2015, pp. 35–41.

[4] P. Fresia and M. Rundo, "Lumped parameter model and experimental tests on a pressure limiter for variable displacement pumps," in *Proc. E3S Web Conf.*, 2020, Paper 07005.

[5] E. E. Paladino, J. A. Lima, R. F. Almeida, and B. W. Assmann, "Computational modeling of the three-dimensional flow in a metallic stator progressing cavity pump," in *Proc. SPE Progressing Cavity Pumps Conf.*, 2008, Paper SPE-114110-MS.

[6] B. Zardin, G. Cillo, M. Rizzoli, and M. Borghi, "Lumped parameter model of vane pumps developed in OpenModelica environment," in *Proc. E3S Web Conf.*, 2021, Paper 05005.

[7] C. Froehlich, W. Kemmetmüller, and A. Kugi, "Model-predictive control of servo-pump driven injection molding machines," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 5, pp. 1665–1680, Sep. 2020.

[8] G. Wang, D. Zhang, C. Li, J. Peng, and X. Zhang, "Research on back pressure control system of injection molding machine based on predictive self-learning optimization PID algorithm," in *Proc. IEEE 7th Int. Conf. Intell. Comput. Signal Process.*, 2022, pp. 949–953.

[9] J. Xu, Z. Ren, S. Xie, Y. Wang, and J. Wang, "Deep learning-based optimal tracking control of flow front position in an injection molding machine," *Optimal Control Appl. Methods*, vol. 44, no. 3, pp. 1–18, 2021.

[10] X. Zhao and A. Vacca, "Analysis of continuous-contact helical gear pumps through numerical modeling and experimental validation," *Mech. Syst. Signal Process.*, vol. 109, pp. 352–378, 2018.

[11] H. Lara-Padilla and C. A. Rodriguez, "Design, simulation and analysis of a positive displacement pump controller for biomedical applications requiring pulsatile flow," in *Proc. IEEE Second Ecuador Tech. Chapters Meeting*, 2017, pp. 1–6.

[12] P. Fisch, M. Holub, and M. Zenobi-Wong, "Improved accuracy and precision of bioprinting through progressive cavity pump-controlled extrusion," *Biofabrication*, vol. 13, 2020, Art. no. 015012.

[13] G. Franchin et al., "Direct ink writing of geopolymeric inks," *J. Eur. Ceram. Soc.*, vol. 37, no. 6, pp. 2481–2489, 2017.

[14] K. A. Woolsey, "Improving progressing-cavity-pump performance through automation and surveillance," *J. Can. Petroleum Technol.*, vol. 51, pp. 74–81, Jan. 2012.

[15] J. Müller, Y. Kouhi, S. Leonow, and M. Mönnigmann, "Modeling pressure pulsation and backflow in progressing cavity pumps with deformable stator," *J. Petroleum Sci. Eng.*, vol. 203, 2021, Art. no. 108402.

[16] E. E. Paladino, J. A. Lima, P. A. Pessoa, and R. F. Almeida, "A computational model for the flow within rigid stator progressing cavity pumps," *J. Petroleum Sci. Eng.*, vol. 78, no. 1, pp. 178–192, 2011.

[17] K. Ogata, *Modern Control Engineering*. Englewood Cliffs, NJ, USA: Prentice Hall, 1997.

[18] E. Todorov and W. Li, "Iterative linear quadratic regulator design for nonlinear biological movement systems," in *Proc. 1st Int. Conf. Inform. Control, Automat. Robot.*, 2004, pp. 222–229.

[19] S. Z. Liu, J. Y. Peng, Y. B. Zhang, and X. B. Chen, "Modeling rotary-screw-driven dispensing systems for electronics packaging," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 4, no. 8, pp. 1359–1365, Aug. 2014.

[20] Y. Liu et al., "On the selection of rheological tests for the prediction of 3D printability," *J. Rheol.*, vol. 67, 2023, Art. no. 791.